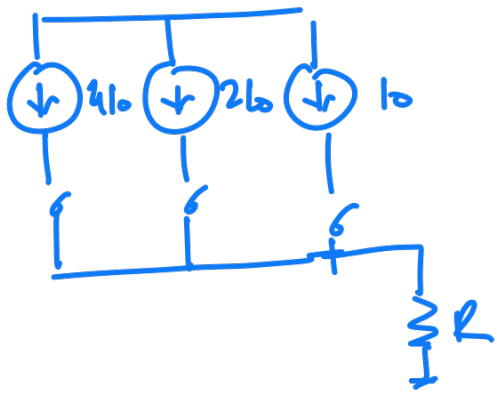
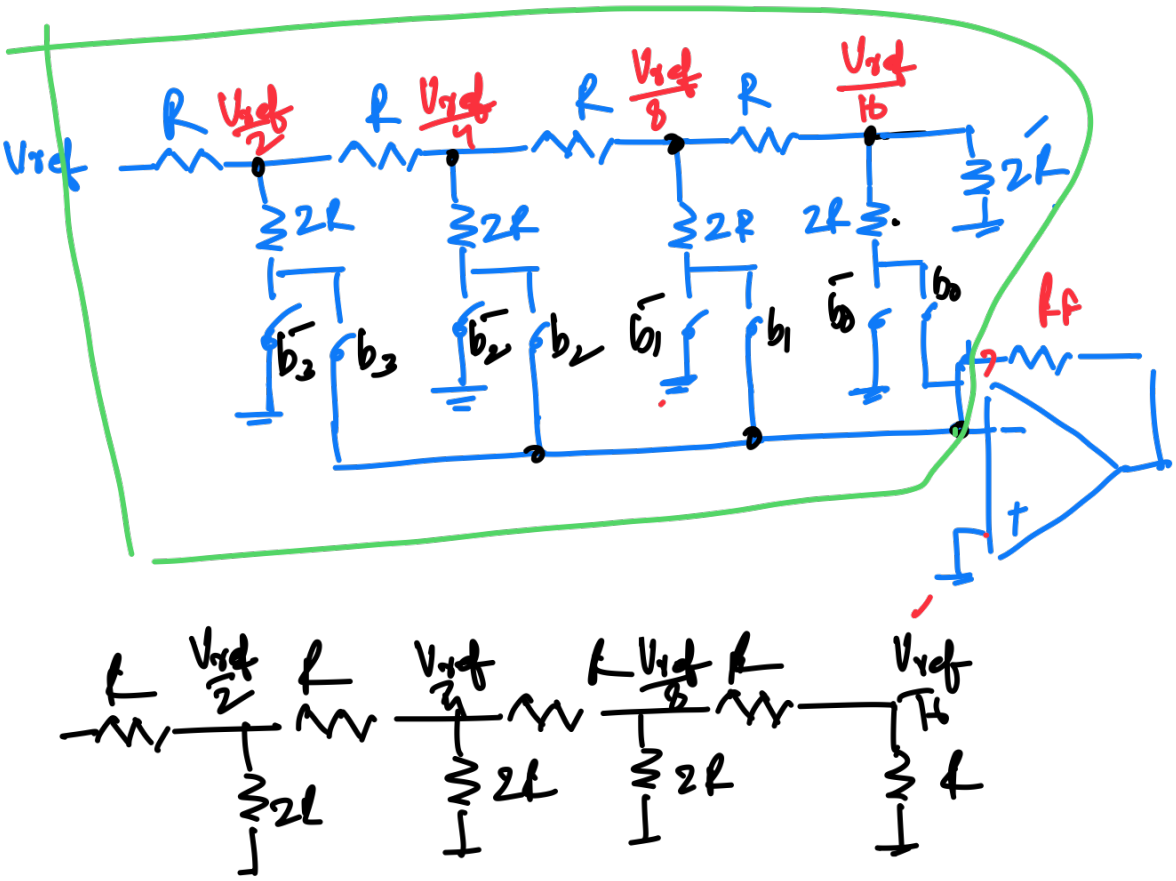
Lecture #40



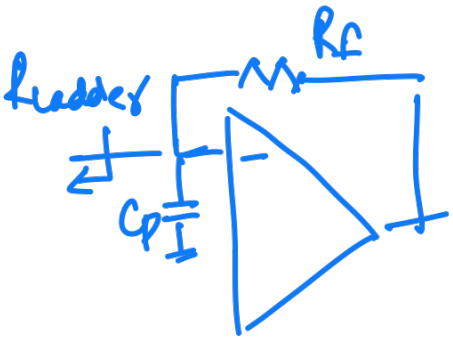


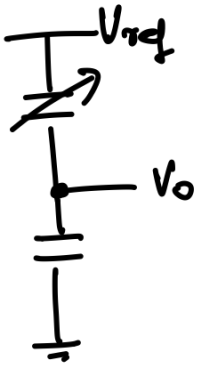


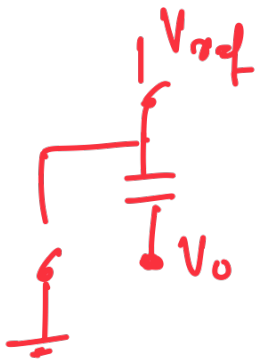
$$I_f = \left[\frac{V_{ref}}{16} \cdot b_0 + \frac{V_{ref}}{8} \cdot b_1 + \frac{V_{ref}}{4} \cdot b_2 + \frac{V_{ref}}{2} \cdot b_3 \right] \times \frac{1}{2R}$$

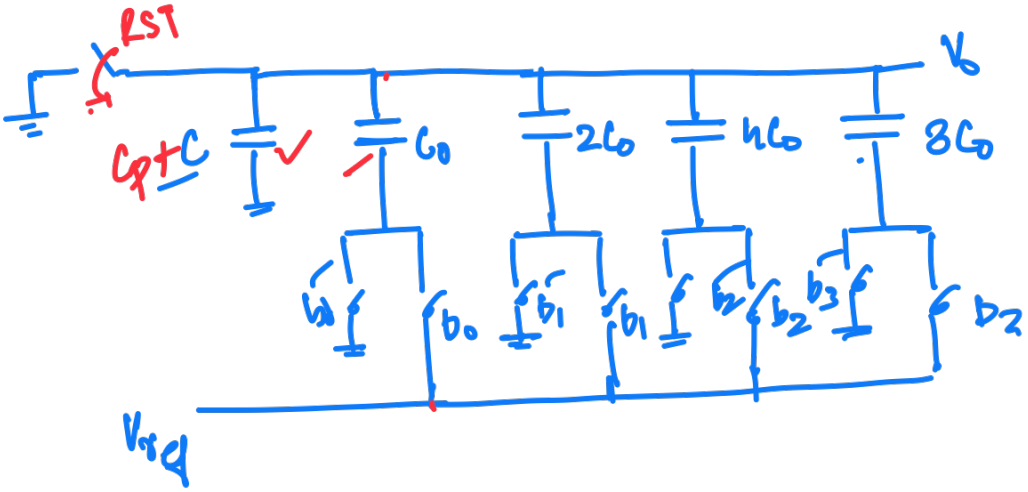
$$V_{out} = -I_f \cdot R_f$$

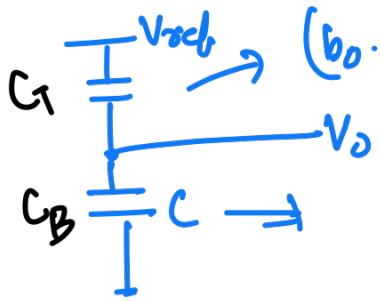
$$= -\frac{V_{ref}}{16 \times 2R} \cdot R_f \left[b_0 + 2b_1 + 2^2 b_2 + 2^3 b_3 \right]$$











$\rightarrow (b_0 \cdot C_0 + b_1 2C_0 + b_2 4C_0 + b_3 8C_0)$

$\rightarrow C + (15C_0 - b_0 C_0 - b_1 2C_0 - b_2 4C_0 - b_3 8C_0)$

$$V_o = \frac{C_T}{C_B + C_T} V_{ref} = \frac{b_0 C_0 + b_1 2C_0 + b_2 4C_0 + b_3 8C_0}{C + 15C_0} \cdot V_{ref}$$

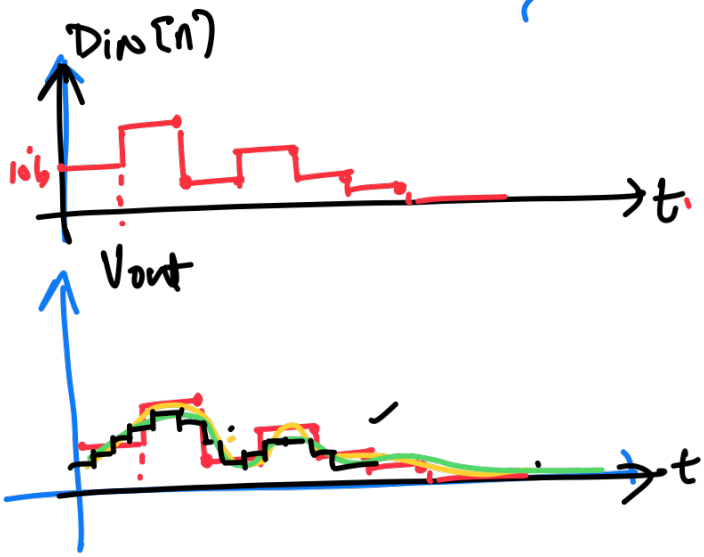
$16C_0$



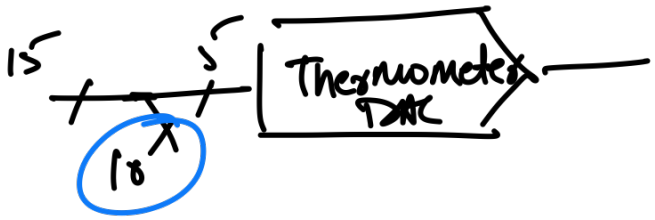
D_{out} (in decimal) $= k V_{in} + Q$.

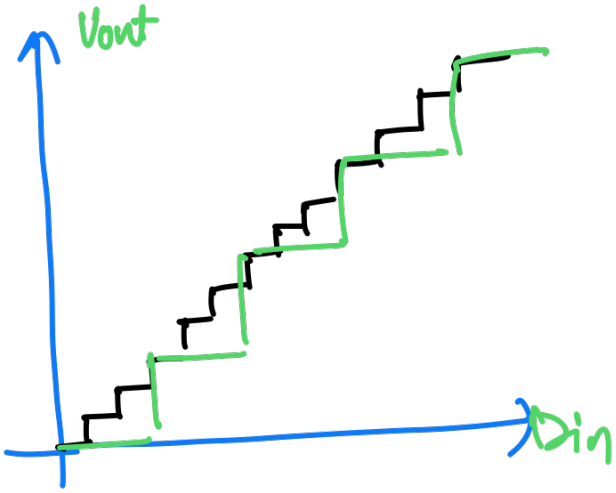


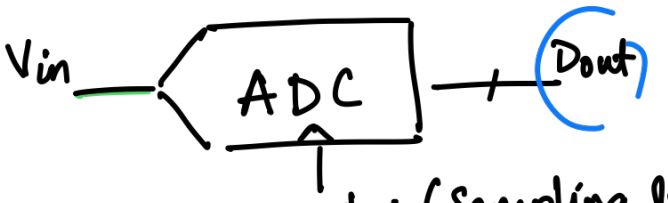
$V_{out} = D_{in} \times \Delta_{LSB}$ (decimal)





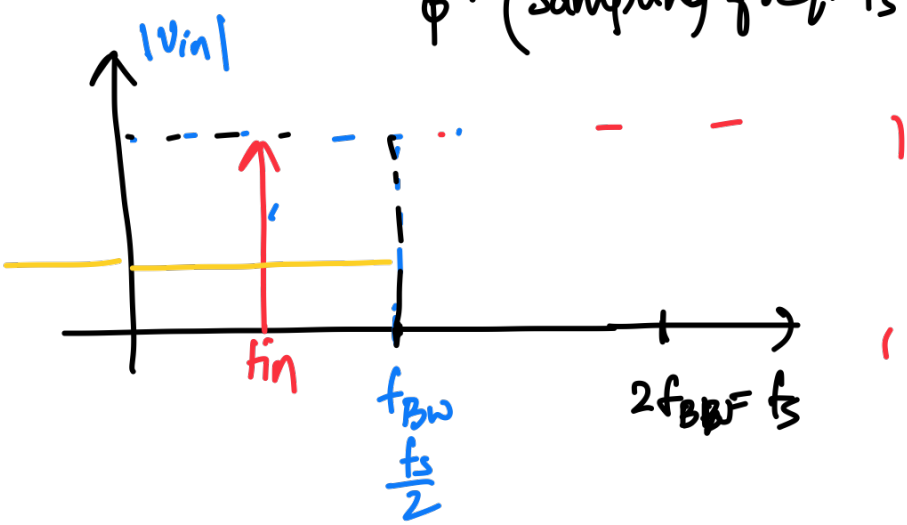






- Nyquist ADC.

ϕ · (Sampling freq. $f_s > 2f_{BW}$)



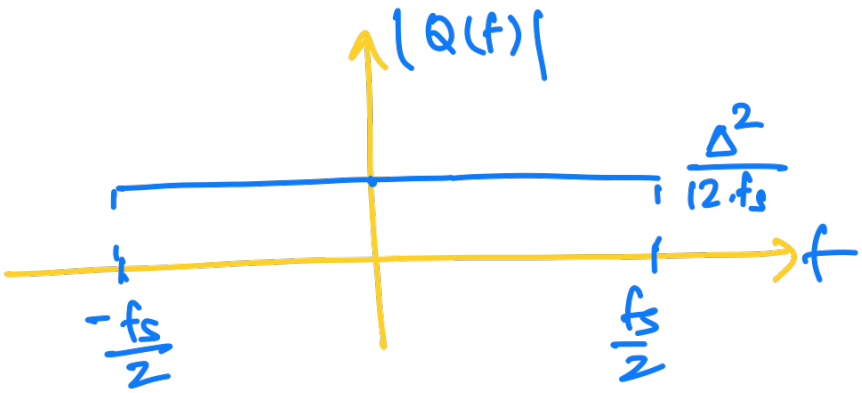
$$D_{out}[nT_s] = k\,V_{in}(nT_s) + q(nT_s)$$

$V_{in}(t) \xrightarrow{\mathcal{F}} V_{in}(f)$

$V_{in}(nT_s) \xrightarrow{\mathcal{F}}$

$q(nT_s) \xrightarrow{\mathcal{F}}$

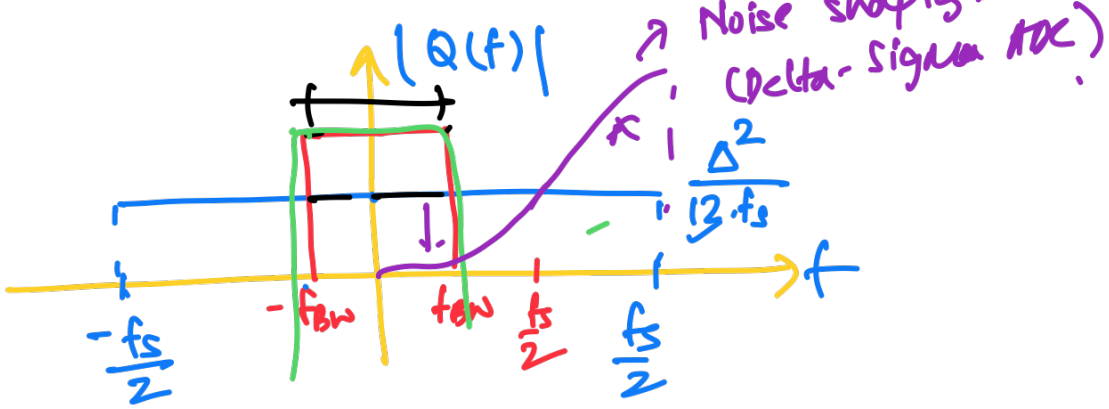
$-\frac{\Delta}{2} < q(n) < \frac{\Delta}{2}$



$E[q] = 0$

$E[q^2] = \frac{\Delta^2}{12}$

$(1-z^{-1})^N$



Noise Power (after filtering) $= \frac{\Delta^2}{12 f_s} \cdot 2 f_{BW}$

$= \frac{\Delta^2}{12} \underbrace{\frac{2 f_{BW}}{f_s}}_{OSR}$

$=$ over sampling ratio

$$SNR = 10 \log_{10} \left(\frac{A^2/2}{\frac{\Delta^2}{12}} \right)$$

$$\downarrow$$

$$10 \log_{10} \left(\frac{A^2/2}{\frac{\Delta^2}{12} \cdot \frac{1}{OSR}} \right)$$